

With agility and adequate partnership strategies towards effective logistics networks

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Abstract

In the mid-1970s, there was a change in many areas of the economy — caused by the laws of supply and demand — from sellers' markets to buyers' markets. This macroeconomic phenomenon — i.e., economics — had decisive consequences for logistics partnerships among companies, and thus for microeconomics. Individual companies today seldom possess all the necessary technological competence to develop and manufacture products of a certain complexity rapidly and economically efficient with advanced technology. Single organizational units no longer handle development and manufacturing. Instead, these tasks are distributed among various companies or among several organizational units within a company. In a logistics network, the companies involved form customer–supplier links. With the exception of the consumer, each customer is a co-producer within the logistics network, and thus is also a supplier. The world of practice reveals that through the course of time, and in dependency upon the concrete demands, different partnership strategies evolve in logistics networks. These may be classified accordingly. We will examine the consequences, which appear gradually, upon the length and intensity of cooperation among companies in a logistics network and derive some plausible future scenarios. For the duration and intensity of cooperation within a logistics network, there are various strategies of partnership. According to supply and demand, as well as the type of product, all of these forms are valid today. The present contribution focuses upon four strategies in particular, namely, the “traditional” customer–supplier relationship, supply management, supply chain management, and virtual organizations. The latter three forms of cooperation were developed in response to short lead times. They are all longer-term in nature. Here, virtual organizations attempt — on the basis of a long-term network — to achieve short-term company cooperation for specific customer orders. Fundamental principles of effective logistics networks concern the agility of a company. Agile competitors, are competitors who understand how to remain competitive by means of proactive amassing of knowledge and competency. Automation with broad implementation of information technology supports agility. © 2000 Elsevier Science B.V. All rights reserved.

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1. The traditional “customer–supplier relationship”

The years prior to 1975 were marked by the post-war boom. There was a heavy demand for all sorts of goods, so a company felt lucky if its needs were met by suppliers. As users, companies took in principle what the supplier had available. Manufacturers had the same problems in purchasing, and this was the case whether they will buy goods (e.g., parts) from a co-distributor or delegate process to an outside co-producer.

This situation changed sometime after 1975. Among others, the increasing industrial productivity and the beginning saturation of some markets led to the unexpected first oil crisis in 1973 to a drastic reduction in demand. This was felt particularly in the investment goods market. As a consequence, the turnaround in power relationships made it possible for the customer — according to the law of supply and demand — to push through price reductions.

With this form of relationship, strategies arise between the producer as a customer and its suppliers, as outlined in Fig. 1.

Price and quality arguments, or productivity in the narrow sense, determine supply and demand. Where friction loss is too high, the customer tends to use insourcing. This is also why there had been a trend in the past towards large and even multinational corporations.

Customer–supplier relationships of this type show low intensity in terms of entrepreneurial cooperation. In principle, the duration of the relationship is indefinite, but in fact is calculated to be short-term: the supplier network is flexible, and any relationship may be replaced with another.

The following definition follows from the strategies outlined in Fig. 1:

The ‘traditional’ *customer–supplier relationship* is defined by the law of supply and demand. Suppliers are chosen on the basis of low prices. Cost reductions are achieved as suppliers play off against each other.

Suppliers can offset the necessity to cut prices by drastically cutting costs themselves. For example, they may keep quality and stockpiling to a minimum. This leads to the following risks:

- reduced quality,
- longer delivery times, and
- poor delivery reliability.

And in fact, these risks were increasingly felt in the early 1980s. Simple consumer goods were not stockpiled and thus ready for delivery, but were produced only upon order. The suppliers’ supply chains broke down and affected the very service level of the customer manufacturer. A new strategy was required.

• Quality:
• The supplier is responsible for meeting the customer's quality specifications. • The customer is responsible for the acceptance and must check the meeting of the specifications.
• Cost:
• The customer chooses a supplier, where quality is sufficient, primarily according to the lowest prices, following the law of supply and demand
• Delivery:
• The customer awards a contract stating desired product, quantity and delivery due date. • Safety stock is necessary in order to avoid the problems caused by delivery delays.
• Flexibility:
• The customer aims for multiple sourcing through finding new suppliers. • If transaction costs become too high, a make-decision is made.
• Relationship between the companies in a logistics network:
• Starting from raw materials and standardized parts, it is the customer who develops all products and processes in the logistics network. • The customer delegates the manufacturing of semi-finished goods or parts of the manufacturing process to suppliers. The customer controls the quality particularly of first deliveries.

Fig. 1. Strategies of the traditional customer–supplier relationship.

2. Through various partnership strategies towards effective logistics networks

According to the situation in terms of supply and demand, as well as the type of product, we found three different partnership strategies. Fig. 2 orders these — as well as the traditional approach — in dependency upon the two dimensions of the “duration” and “intensity” of cooperation.

These two dimensions proved to be very important features of a characteristic of logistics networks. During time, partnership strategies evolved first, from short to long duration, then to long duration and high intensity. Actually, Supply Chain Management is a strategy to handle very important volumes of business. During the last years, the concept of a virtual organization allows a highly intensive cooperation during a short time. As a virtual organization is generally built from an existing association in the form of a network of long-term duration, it can be said that the term *virtual organization* includes implicitly the long-term aspect.

The three different partnership strategies will now be presented in detail. A comprehensive discussion can also be found in [1] and [2], chapter 2.

2.1. Supply management

The term “supply management” originated, along with the concept of “just-in-time,” in the 1980s. It stands for an approach to supply and demand that functions not only according to price and quality. For delivery, unreliability on the part of the suppliers

result in opportunity costs for the manufacturer, if it cannot — in turn — supply its own customers.

Shorter delivery times, however, can only be achieved by means of close cooperation with the supplier. This is a crucial recognition that — of necessity — must lead in a logistics network to a change in the form of cooperation between customer (user or manufacturer) and supplier (in its role as a co-distributor or co-producer). This new behavior is based on strategies shown in Fig. 3.

This type of cooperation with suppliers demands extensive preparation. For this reason, such long-term relationships cannot be established and maintained with many partners. The above strategies result in the following definition:

Supply management is a strategic and long-term reduction of the number of suppliers to achieve fast and easy operational order servicing. The choice of a supplier is made in view of total costs, i.e., under consideration of all opportunity costs.

Supply management, in short-term order servicing, leads to the elimination or reduction of friction loss caused by order negotiations. With this, many of the advantages of company-internal production for fast throughput time can be retained.

Customer–supplier relationships of this type show, however, low intensity in terms of entrepreneurial cooperation. The relationships have to be checked again and again with regard to their validity. The following particular risks may result:

- wrong choices when reducing the number of suppliers;
- changes in crucial conditions on the part of suppliers; and
- unexpected shift to a sellers’ market.

2.2. Supply chain management

The increasing competence of workers and staff in the command of technologies and operations led, as early as the 1980s, to tendency towards profit centers. The broad application of supply management, together with a strong buyers’ market, resulted, in the early 1990s, in increased demands upon internal organizational units. If transaction costs

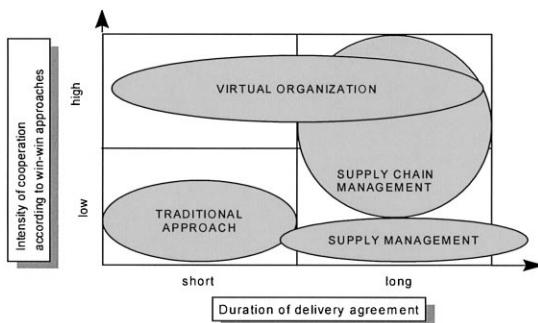


Fig. 2. Ordering of various strategies of cooperation in a logistics network.

• Quality:
• The supplier achieves a minimum level of quality (according to its own quality control, external certification). • In order to control the quality of the supplier, the customer has access to its production facilities. Both parties mutually improve quality in a logistics network.
• Cost:
• Through single sourcing greater business volume and thus lower cost prices are achieved. • (Long-term) blanket orders allow intermediate stores to be reduced. • The choice of a supplier is made according to total costs, that is, in consideration of opportunity costs.
• Delivery:
• (Long-term) blanket orders reduce total throughput time (supplier and customer). • There is now direct delivery on demand to the production facilities of the manufacturer.
• Flexibility:
• Single sourcing also provides for a replacement supplier. • As to the rest, the buyers' market secures the robustness of the approach: transaction costs are small, and it is relatively easy to secure a replacement supplier (buy-decision).
• Entrepreneurial cooperation in the logistics network:
• Demands on products and processes to be delivered are mutually defined. • The supplier is consulted about each (further) development.

Fig. 3. Strategies of supply management.

decrease, there are ever fewer reasons for keeping parts of the value-added chain within the company, particularly if processes with no core competency are involved. The result is outsourcing.

At the same time — again due to the strengthening buyers' market — the demand for short product innovation times ("time to market") is more easily fulfilled. Whenever company-internal departments cannot make the necessary transitions — i.e., when the company's own organization becomes slow and expensive — trans-corporate product and process development with co-producers is shown to be advantageous. When product development becomes more and more costly, entrepreneurial risk may then be more widely distributed.

Reducing the time for product innovation demands intensification of entrepreneurial cooperation with co-producers. This will be true for all levels of the logistics network, thus the term "supply chain" management. Good communication paths, both technical (telephone, fax, ISDN and EDI) and personal (regular meetings at all hierarchical levels) are necessary. The reader is referred here to Alberti [3].

Reduced time for product innovation is often linked with unpredictable demand. To avoid stock-outs or obsolete inventory, short lead times are an important feature. This is only possible through intensive cooperation of all the companies involved in the value-adding chain (compare Fisher, 1997) [7].

In order to support the comprehensive requirements in planning and control in a supply chain management concept, a specific software called SCM software has been developed in recent years and continues to be developed intensively. As it is the case for coping with many other recent issues in modern management, supply chain management requires organizational innovation, as well as IT innovation.

Fig. 4 sums up the strategies. They are to be understood as complementary to the strategies shown in Section 2.1.

The focus is again on operating efficiency criteria. Resources must be implemented in the best possible manner.

This type of cooperation gives co-producers insight into the participating companies. One absolute prerequisite is the long-term formation of trust. Entrepreneurial cooperation thus becomes intensive. The above strategies result in the following definition:

Supply chain management is the coordination of strategic and long-term cooperation among co-producers in the total logistics network for the development and production of products, both in production and procurement and in product and process innovation. Each co-producer is active within its own area of core competence. The choice of co-producer is made with chief impor-

• Quality:
• Each co-producer <i>feels responsible</i> for the satisfaction of the end user. • Quality requirements are developed and improved mutually.
• Cost:
• All advantages of supply management are maintained. This leads generally to lower transaction costs. • Sharing of methods and know-how among co-producers reduces costs. • Each co-producer is active in its area of core competence. This yields the best possible return from the resources implemented (including time).
• Delivery:
• The same logistics are necessary for all co-producers (same operational procedures, documents, etc.). • Planning and control systems are linked (for example, via EDI). • The choice of co-producer depends with chief importance upon speed, that is, the co-producer's contribution to short lead times.
• Flexibility:
• All co-producers give impetus towards product development. • Once again, the buyers' market guarantees that the approach be robust: transaction costs are low, and replacement suppliers may be arranged relatively easily (buy decision).
• Entrepreneurial cooperation in the logistics network:
• All co-producers are involved in product and process development from the start. • All co-producers are involved in planning and control.

Fig. 4. Strategies of supply chain management.

tance according to its potential towards realization of short lead times.

Supply chain management, in short-term order servicing, results in the elimination or reduction of friction loss that otherwise results from procurement negotiations. In principle, the advantages of a profit center organization are carried over to independent companies.

On the other hand, profit center organizations can significantly improve, through the simple application of supply chain management, the efficiency of their internal logistics networks. The decisive factor will be the degree of entrepreneurial thinking and action.

This type of close relationship also carries the possibility of *risks* such as:

- Abuse of the knowledge gained from cooperation with co-producers in order to enter business relationships with their competitors; and
- Investiture by co-producers which — due to too brief cooperation periods — is not profitable.

2.3. Virtual organizations

Is there any possible form of temporally restricted and yet intensive cooperation, such as for non-repe-

titive products that solve a customer's specific problem? Virtual organizations are a potential answer. The reader is referred here to Davidow and Malone [6] and Goldman et al. [9].

The adjective *virtual* means, according to Merriam-Webster [8], 'possessed of certain physical virtues.' In reference to the business world, this means that a company functions as such, even though it is not a company in a legal sense.

The concept of virtuality aims to utilize the advantages of supply chain management as soon as the customer defines its individual needs. In order to fulfill those needs, several co-producers — or departments of a company — must join together. To the customer, they stand as a single company, but later they will again separate. These same departments may then join other companies to form new virtual organizations.

The strength of virtual organizations lies in their ability to form quickly. In the world of practice, co-producers must already be familiar with each other. Fig. 5, taken from Bruetsch et al. [5], illustrates this concept.

Independent organizations of various types strive to achieve an association in the form of a network of *long-term duration* (indicated in Fig. 5 by thin arrows), so that each partner gains competitive advan-

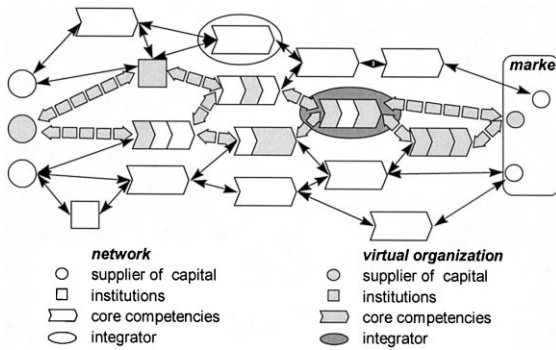


Fig. 5. The virtual organization and underlying network of potential partners.

tages. Possible hindrances to cooperation have to be dispelled in this phase in order for a relationship of trust to develop among the individual network participants. This requires — just as in supply chain management — the establishment of good paths of communication of both a technical and personal nature. In some cases of development cooperation, contractual agreements make sense.

Fig. 6 gives an overview of the strategies. They are to be seen as complementary to the strategies discussed in Section 2.2.

Of enterprise efficiency criteria, a company's flexibility is particularly important here. In addition, in order to form virtual organizations rapidly, the

company boundaries of the potential co-producers in the network must already be open. In this way, entrepreneurial cooperation can be very intensive. Again, as an absolute prerequisite, trust must develop long-term. As a general principle, competition within the network is usually ruled out.

The above strategies result in the following definition:

A *virtual organization* is a short-term form of cooperation among legally independent co-producers in a logistics network of long-term duration of potential business partners for the development and manufacturing of a product. This is true for procurement and production, as well as for product and process innovation. Co-producers produce the service on the basis of mutual values and act towards the third party as a single organization. Each co-producer is active within the area of its core competence. The choice of co-producer depends upon the co-producer's innovative power and its flexibility to act as a partner in the logistics network.

In short-term order servicing, the virtual organization, like supply chain management, results in the entire logistics network in elimination or reduction of friction loss that can occur due to procurement negotiations.

• Quality: • Each co-company carries <i>extensive responsibility</i> for end-user satisfaction. • Action guidelines, structures and processes of the virtual organization are developed mutually, as is the basic network of potential partners.
• Cost: • All advantages of supply chain management are maintained. This leads to lowest costs.
• Delivery: • The logistics network for a specific order is formed rapidly. • The same operational procedures, documents, etc. are prerequisites. • Identical information systems allow a maximal exchange of information during mutual product development and production.
• Flexibility: • Criteria for the choice of a co-enterprise are 1) its flexibility to enter as a partner into a logistics network 2) its innovative power, that is, its flexibility in creating customer value and 3) the extent of shared value orientations.
• Entrepreneurial Cooperation in the logistics network: • All potential partners form a long-term network. One has the role of a broker, in order to put together the virtual organization according to a concrete demand. • All co-producers supply product and process development and planning and control from the start. They share mutual involvement and responsibility for success or failure.

Fig. 6. Strategies of a virtual organization.

A broker is required for the rapid formation of networks. In the case of non-repetitive production, the broker often serves also as a center for order processing, i.e., for planning and control. If throughput time must be very short, the planning autonomy of the companies involved must be curtailed. A virtual organization then takes on, in terms of logistics requirements, the characteristics of a centrally managed cost center.

The other way around, cost center organizations can increase the efficiency of their internal logistics networks substantially by the simple application of the principles of the virtual organization. The decisive factor here is the degree of flexibility of the cost center to contribute to the goals of the total enterprise.

Virtual organizations involve *risks*. In addition to those outlined for supply chain management, virtual organization can entail the following hazards:

- Lack of competition with regard to potential partners in the network means that certain orders cannot be taken on;
- Legal problems (loss and gain distribution, copyrights and rights of ownership); and
- The volume of business is too small to justify the long-term expense involved.

To reduce the risk of a lack of business volume, each of the partners must attempt to anticipate the customer's needs. This demand on agile companies requires study of the actual use of products in order to develop proactive proposals for the implementation of new products, which have not even occurred to the customer.

3. From lean companies to agile companies: fundamentals of effective logistics networks

APICS [4] defines *lean production* as the minimization of all required resources (including time) for the various activities of the company. It involves identifying all non-value-adding activities in development and production within the entire logistics network, extending to and including the consumer as *waste* and eliminates them.

Lean companies are companies that apply the principles of lean production to all areas within the organization.

Since being introduced, the philosophy of lean production of Womack et al. [11] has often been taken to extremes. It served as a convenient justification for firing and not replacing staff members. Some people postulated polemically that the contradictory goals of the company, as outlined in Section 1.2.2, could be resolved. They did not consider explicitly the target area of flexibility, for its goals are usually long-term in nature. The customer does not readily recognize that the building of such competencies is value-adding.

Today there is a wider recognition that the value-adding process may not be evaluated from the perspective of the customer alone, who generally is concerned only about his temporally limited order. Too often, management aiming for short-term success simply dissolved long-term competency, such as concealed assets, in order to demonstrate short-term gains. This familiar but irresponsible action led to company anorexia and resulting paralysis — at a time when company agility was required.

Agility is defined in Merriam-Webster [8] as ability to move, dexterity. Agility, according to Goldman et al. [9], is the quality or state of being agile.

If businesses are challenged by new competitive drivers that are changing fundamentally the way business is done, agility is a must for a company in order to survive, and even more, to compete. In other words, ability allows to gain advantage from a competitive environment where insecurity dominates, and which differs structurally from mass production.

Agile manufacturing refers to the building up of potentials or scope (or 'play') in the right place at the right time and in the right amount.

Agile companies are companies that apply the principles of agile manufacturing to all areas in their organizations.

Agile competitors, according to Preiss et al. [10], are competitors who understand how to remain competitive by means of *proactive* amassing of knowledge and competency.

The crucial factor in a company's agility is thus knowledge — knowledge that was lost in lean companies. Agile companies are deliberately not lean companies. Instead, they develop potentials and scope that the customer does not see (and thus does not recognize as value-adding). Here are some examples.

- Short-term: formation of multiple parallel executions of orders and running order coordination by the different partners in the logistics network linked by pull logistics; establishment of overlapping activities in part-processes to coordinate push logistics. Each of these makes possible rapid business processes of high quality.

- Medium-range to long-term: building staff capable of qualitatively flexible work assignments by means of training qualifications and coordination in groups; setting up a production infrastructure that may be implemented flexibly. Both measures lead to flexibility in the implementation of resources.

- Medium-range: building either overcapacity, or quantitatively flexible capacity, and/or goods in stock. Either of these measures allows a response to unplanned demand or shifts in demand with short delivery times. In the production of capital goods, capacity measures that reduce ordering deadlines are given preference.

- Medium-range: development of competency in proactive service. The producer gathers information on the product in the user phase. By evaluating this information, the maker is in a position to recognize changes in customer demands. The maker may — proactively — offer an upgrade or new product, even before the customer is aware of his needs. In this way, he sells to the customer a solution rather than a single product. The customer feels cared for (“total care”).

- Long-term: development of knowledge and methods to develop and manufacture products in manifold variants. This knowledge allows flexibility in reaching customer value. The maker is able to give positive answers to customer requests at crucial moments, which increases the ratio of successful bids.

- Long-term: development of knowledge and methods in reconfiguring the organization. According to the product, departments take on a new structure and work in cooperation with other organizations. This knowledge allows the flexibility to enter

as a partner into a logistics network. At just the right moment, a company can immediately become a partner in a logistics network.

4. Supporting agility by information technology

Automation with broad implementation of information technology supports agility. Telefax and electronic data interchange are examples of basic technologies allowing worldwide direct transfer of data. A concrete case is the electronic data exchange for administration, commerce and transport (Edifact) interface. But the Internet and corresponding company-internal Intranets are useful. Here are some examples.

4.1. Multiple parallel order execution by several independent partners in the logistics network

Electronic mail (E-mail) over the Internet is used for worldwide order coordination. One distribution mailbox can store all addresses of those involved in the logistics network. Informal inquiries to this mailbox as to order status — and answers — are thus immediately available to all partners in the logistics network.

4.2. Running order coordination

Transport companies supply the customer with information via the World Wide Web on the exact location of their goods. This occurs in the ideal case through self-identification of goods by means of attached transponders.

4.3. Proactive service

Automobile manufacturers have access to the product and service data bank of their customers via service centers. Their own data banks then allow effective evaluation of the “life data” of the product during the user phase.

4.4. Competency to develop and produce products in manifold variants

In jeans or shoe stores, customer measurements are taken and transmitted directly to production

workshops. A few days or weeks later, the customer receives the finished, made-to-order product. Or, a customer may configure a specific insurance policy directly on the home page of an insurance company. He types in the desired parameters, which he then may vary according to need.

4.5. Competency in reconfiguring the organization

Partners within a specific logistics network transmit order data using mutually configured software. The data are used directly to control processes (such as in machines). In the manufacturing of cars or airplanes, the partners represent their business objects in a standardized form. To this purpose, they long ago developed special standards of EDI among others IGES (later STEP, for engineering purposes) and Edifact with its variations (e.g., Odette in the automotive industry) for sales order processing.

Increasingly, the predominant consumer market demands agile competitors and individualized production even of consumer goods. What the consumer buys is more solutions and values than pre-defined functions, whereby the product is ever more frequently defined only in direct interaction with the customer. The behavior of the consumer is more spontaneous and less predictable. Clearly identifiable market segments will disappear, and brand names will serve increasingly to reflect the personality of the customer instead of providing a function as they used to do.

5. Conclusion

For the duration and intensity of cooperation within a logistics network, there are various strategies of partnership. According to supply and demand, as well as the type of product, all of these forms are valid today. This chapter has focused upon four strategies in particular, namely the “traditional” customer–supplier relationship, supply management, supply chain management, and virtual organizations. The latter three forms of cooperation arose in response to short lead times. They are all longer-term

in nature. Here, virtual organizations attempt to — on the basis of a long-term network — achieve short-term company cooperation for specific customer orders.

An important fundamental of effective logistics networks is the agility of the companies involved. Agile companies are not lean companies. Rather, they are able to remain competitive by means of the proactive development of knowledge and competence. This development is, in the shortsighted view of the customer, not value-adding. Timely implementation of appropriate temporal, local, and quantitative potentials and degrees of freedom will be competitively decisive. Agility is supported by the use of information technology, in particular EDI, such as the Internet.

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